Question: Why is $\text{rank}(A) \overset{(3)}{=} \text{rank}(A^TA) = \text{rank}(AA^T) \overset{(3')}{=} \text{rank}(A^T)$ (D)

(D) $\text{rank}(A) = \text{rank}(A^T)$.

$\dim(\text{Row}(A)) = \dim(\text{Col}(A^T))$

(3') follows from (3) applying to A^T.

MA1522 Linear Algebra for Computing

Lesson 22: Revision of Chapters 3 and 4

(2) $\text{rank}(A) = \text{rank}(A^TA)$

(2-1) $\text{Null}(A) = \text{Null}(A^TA)$.

$\left(Av = 0 \iff Av \cdot Av = 0 \iff v^T(A^TAv) = 0 \right)$ Lesson x problem y.

A: $m \times n$ $\quad A^TA$: $n \times n$

$n = \text{rank}(A) + \text{nullity}(A)$ (2-1) $\Rightarrow$ $\text{rank}(A) = \text{rank}(A^TA)$

$= \text{rank}(A^TA) + \text{nullity}(A^TA)$

In SVD we find $\{v_1, \dots, v_r\}$ eigenvectors of A^TA with eigenvalue $\sigma_i^2 \neq 0$.

$u_i := \frac{1}{\sigma_i} Av_i$.

Question: Why $\{u_i\}$ is orthonormal?

$u_i \cdot u_j = \frac{1}{\sigma_i \sigma_j} v_i^T A^TA v_j = \frac{1}{\sigma_i \sigma_j} v_i^T (\sigma_j^2 v_j) = \frac{\sigma_j}{\sigma_i} v_i \cdot v_j = \begin{cases} 1 & \text{if } i=j \\ 0 & \text{o.w.} \end{cases}$

m $\{$ (n)

Announcements

bring calculator, dark pencil. student ID card.

- **Final Exam**: **Tuesday 24 November 2026, 5:00 pm**. Closed book; one helpsheet is permitted. See the **Instruction of Final Exam** file on Canvas.

- Please do the **Student Feedback Exercise**.

- Consultation 2:00 – 3:00 pm 19, 20 Nov. S16-05-18 (subject to change).

Submission

- You only need to submit your attempts at those labelled **Question 1** to **Question 3**.
- If you have written your answer on A4 paper, scan or take pictures of your work (make sure the images can be read clearly). Merge all your images into one PDF file, in page order.
- If you have written or typed your answer digitally, print it into a PDF file.
- Name the PDF file by `StudentNo_lesson22` (e.g. `A1234567Z_lesson22.pdf`).
- Submit the PDF file to Canvas assignments, **Participation(L1): Lesson 22**.
- Deadline: today, **12:30 PM**.
- If you do not know how to upload your submission, come to ask me after class.

Pollev



Key definitions: Chapters 3 and 4

Chapter 3		
Concept	**Definition**	**Algorithm**
span(S)		$\times$
$\mathbf{v}$ is in the span of S		
S spans $\mathbb{R}^n$		
span(T) $\subseteq$ span(S)		
V is a subspace of $\mathbb{R}^n$		$\times$
The solution space of $A\mathbf{x} = \mathbf{0}$		
The solution set of $A\mathbf{x} = \mathbf{b}$ as an affine space		
S is linearly independent		
S is a basis for V		
Extend S to a basis for V		
$[\mathbf{v}]_S$, the coordinate vector of $\mathbf{v}$ relative to S		
dim(V)		
The transition matrix from T to S		

Chapter 4		
Concept	**Definition**	**Algorithm**
Row(A)		
Col(A)		
Null(A)		
rank(A)		
nullity(A)		
A is of full rank		
A is invertible		

Here S, T are subsets of $\mathbb{R}^n$, V is a subspace of $\mathbb{R}^n$, $\mathbf{v} \in \mathbb{R}^n$, and A is an $m \times n$ matrix.

Warning

1. This is not necessarily an exhaustive list.
2. You should also know their properties.

Question 1

(In my opinion, this is slightly more difficult than the problems in final exam). I will start to solve it. 10:30.

Let a be a real number, and let

$$U = \{ (x, y)^T \in \mathbb{R}^2 \mid x^2 + 2axy + y^2 = 0 \}.$$

1. Find all values of a for which U is a subspace of $\mathbb{R}^2$.

2. For each such value of a, find a basis for U and $\dim(U)$.

(Hint: $x^2 + 2axy + y^2 = (x + ay)^2 + (1 - a^2)\, y^2$.)

Question 1 — solution: part 1

$$U = \left\{ \begin{pmatrix} x \\ y \end{pmatrix} \in \mathbb{R}^2 \;\middle|\; x^2 + 2axy + y^2 = 0 \right\}.$$

$$x^2 + 2axy + y^2 = (x+ay)^2 + (1-a^2)y^2$$

- Step 1. solve $(x+ay)^2 + (1-a^2)y^2 = 0$

$$(x+ay)^2 = (a^2-1)y^2.$$

① $x+ay = \pm\sqrt{(a^2-1)y^2}$

② $x = -ay \pm |y|\sqrt{a^2-1}$.

Step 2. Discuss when can we take a square root?

Case 1: $a^2-1<0 \iff |a|<1$.

① $\sqrt{(a^2-1)y^2}$ has to be zero. Namely $y=0$.

Then it becomes $x+a\cdot 0 = 0 \implies x=0$.

So if $|a|<1$, then $U = \left\{ \begin{pmatrix} 0 \\ 0 \end{pmatrix} \right\}$ is a subspace $\begin{cases} \text{basis} = \emptyset \text{ (empty set)} \\ \dim(U) = 0. \end{cases}$

Case 2: $a^2-1 \geq 0 \iff a = 1$ or -1.

① becomes $x+ay=0$.

$$U = \left\{ \begin{pmatrix} x \\ y \end{pmatrix} \;\middle|\; x+ay=0 \right\} \text{ is a subspace.}$$

- $a=1$. $U = \operatorname{span}\left\{ \begin{pmatrix} 1 \\ -1 \end{pmatrix} \right\}$. $\dim(U) = 1$.
- $a=-1$. $U = \operatorname{span}\left\{ \begin{pmatrix} 1 \\ 1 \end{pmatrix} \right\}$

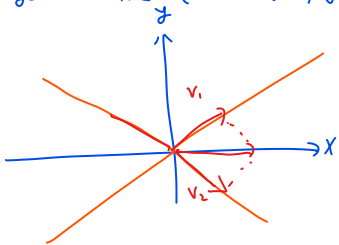
Question 1 — solution: part 2

Case 3: $a^2-1 > 0 \iff |a| > 1$.

1) $x + ay = \pm\sqrt{(a^2-1)y^2}$.

$$x = -ay \pm |y|\sqrt{a^2-1}.$$

- $y \geq 0$: $x = (-a \pm \sqrt{a^2-1})\, y$.
- $y \leq 0$: $x = (-a \mp \sqrt{a^2-1})\, y$.



$$U = \left\{ \begin{pmatrix} x \\ y \end{pmatrix} \;\middle|\; x = (-a+\sqrt{a^2-1})\, y \right\}$$

union with

$$\left\{ \begin{pmatrix} x \\ y \end{pmatrix} \;\middle|\; x = (-a-\sqrt{a^2-1})\, y \right\}.$$

Try $a = 2$, $v_1 = \begin{pmatrix} -2+\sqrt{3} \\ 1 \end{pmatrix}$, $v_2 = \begin{pmatrix} -2-\sqrt{3} \\ 1 \end{pmatrix}$

$$v_1 + v_2 = \begin{pmatrix} -4 \\ 2 \end{pmatrix} \notin U.$$

you can find $v_1, v_2 \in U$ but

$v_1 + v_2 \notin U$.

Question 2

Let a be a real number, and let

$$\mathbf{s}_1 = \begin{pmatrix} 1 \\ 0 \\ 2 \\ 1 \end{pmatrix}, \ \mathbf{s}_2 = \begin{pmatrix} 0 \\ 1 \\ -1 \\ 1 \end{pmatrix}, \ \mathbf{s}_3 = \begin{pmatrix} 1 \\ 1 \\ 0 \\ 3 \end{pmatrix}, \qquad \mathbf{t}_1 = \begin{pmatrix} 1 \\ 2 \\ 0 \\ 3 \end{pmatrix}, \ \mathbf{t}_2 = \begin{pmatrix} 3 \\ 4 \\ 1 \\ 8 \end{pmatrix}, \ \mathbf{t}_3 = \begin{pmatrix} 3 \\ a+2 \\ 2-a \\ a+7 \end{pmatrix}.$$

Let $V = \text{span}\{\mathbf{s}_1, \mathbf{s}_2, \mathbf{s}_3\}$; then $S = \{\mathbf{s}_1, \mathbf{s}_2, \mathbf{s}_3\}$ is a basis for V. Let $T = \{\mathbf{t}_1, \mathbf{t}_2, \mathbf{t}_3\}$.

1. Find all values of a for which $\mathbf{t}_1, \mathbf{t}_2, \mathbf{t}_3$ all lie in V.

2. Find all values of a for which T is a basis for V.

3. Let $a = 1$. Find the transition matrix from S to T.

Question 2 — solution: part 1

$$\left(\begin{array}{ccc|ccc} 1 & 0 & 1 & 1 & 3 & 3 \\ 0 & 1 & 1 & 2 & 4 & a+2 \\ 2 & -1 & 0 & 0 & 1 & 2-a \\ 1 & 1 & 3 & 3 & 8 & a+7 \end{array}\right) \quad \begin{matrix} \\ \\ = s_1 \\ = s_2 \\ = s_3 \end{matrix} \quad \begin{matrix} \\ \\ = t_1 \\ = t_2 \\ = t_3 \end{matrix}$$

$R_3 - 2R_1$

$R_4 - R_1$

$$\left(\begin{array}{ccc|ccc} 1 & 0 & 1 & 1 & 3 & 3 \\ 0 & 1 & 1 & 2 & 4 & a+2 \\ 0 & -1 & -2 & -2 & -5 & -4-a \\ 0 & 1 & 2 & 2 & 5 & a+4 \end{array}\right)$$

$R_3 + R_2$, $R_4 - R_1$

$$\left(\begin{array}{ccc|ccc} 1 & 0 & 1 & 1 & 3 & 3 \\ 0 & 1 & 1 & 2 & 4 & a+2 \\ 0 & 0 & -1 & 0 & -1 & -2 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right)$$

these are always zero

$R_4 + R_3$

$$\left(\begin{array}{ccc|ccc} 1 & 0 & 1 & 1 & 3 & 3 \\ 0 & 1 & 1 & 2 & 4 & a+2 \\ 0 & 0 & -1 & 0 & -1 & -2 \\ 0 & 0 & 1 & 0 & 1 & 2 \end{array}\right)$$

So for all real number a, $t_1, t_2, t_3 \in V$.

Question 2 — solution: part 2

$$\left(S_1\ S_2\ S_3 \middle|\ t_1\ t_2\ t_3\right) \xrightarrow{\text{row operations}} \left(\begin{array}{ccc|ccc} 1 & 0 & 1 & 1 & 3 & 3 \\ 0 & 1 & 1 & 2 & 4 & a+2 \\ 0 & 0 & -1 & 0 & -1 & -2 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right) \xrightarrow[R_3 \leftrightarrow -R_3]{\substack{R_1+R_3 \\ R_2+R_3}} \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 2 & 1 \\ 0 & 1 & 0 & 2 & 3 & a \\ 0 & 0 & 1 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right) \quad P_{T \to S}$$

Question: For which a, t_1, t_2, t_3 forms a basis of V?

since these are 3 vectors inside V with $\dim(V)=3$, it is equivalent to check whether $\{t_1, t_2, t_3\}$ is l.I.

$$[t_1]_S = \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix}, \quad [t_2]_S = \begin{pmatrix} 2 \\ 3 \\ 1 \end{pmatrix}, \quad [t_3]_S = \begin{pmatrix} 1 \\ a \\ 2 \end{pmatrix}.$$

Now that $\{t_1, t_2, t_3\}$ is linearly independent $\iff \{[t_1]_S, [t_2]_S, [t_3]_S\}$ is linearly independent.

$$\iff \det \begin{pmatrix} 1 & 2 & 1 \\ 2 & 3 & a \\ 0 & 1 & 2 \end{pmatrix} \neq 0.$$

Now $\det \begin{pmatrix} 1 & 2 & 1 \\ 2 & 3 & a \\ 0 & 1 & 2 \end{pmatrix} \overset{R_2 - 2R_1}{=} \det \begin{pmatrix} 1 & 2 & 1 \\ 0 & -1 & a-2 \\ 0 & 1 & 2 \end{pmatrix} = \det \begin{pmatrix} -1 & a-2 \\ 1 & 2 \end{pmatrix} = -2 - (a-2)$

$$= -a$$

So $\{t_1, t_2, t_3\}$ forms a basis of $V \iff a \neq 0$.

Question 2 — solution: part 3

For $a=1$, find $P_{S\to T}$.

$$P_{T\to S} = \left([t_1]_S \;\; [t_2]_S \;\; [t_3]_S \right) = \begin{pmatrix} 1 & 2 & 1 \\ 2 & 3 & 1 \\ 0 & 1 & 2 \end{pmatrix}$$

$$P_{S\to T} = \left(P_{T\to S} \right)^{-1} = \begin{pmatrix} -5 & 3 & 1 \\ 4 & -2 & -1 \\ -2 & 1 & 1 \end{pmatrix}$$

by calculator

Question 3

Recall rref preserves { the row space; the linear combination/relation of columns.

Let c be a real number, and let A be a 5×4 matrix with rows $\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3, \mathbf{a}_4, \mathbf{a}_5$ such that

$$\text{rref}(A) = \begin{pmatrix} 1 & 0 & c & 0 \\ 0 & 1 & -2 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \qquad \text{rref}(A^T) = \begin{pmatrix} 1 & 0 & 2 & 0 & -1 \\ 0 & 1 & c & 0 & 0 \\ 0 & 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}.$$

This gives a basis of Row(A)

pivot columns

$A = \begin{pmatrix} - & a_1 & - \\ & \vdots & \\ - & a_5 & - \end{pmatrix}$, $A^T = \begin{pmatrix} | & & | \\ a_1^T & \cdots & a_5^T \\ | & & | \end{pmatrix}$

1. Find all values of c for which $\{\mathbf{a}_3, \mathbf{a}_4, \mathbf{a}_5\}$ is a basis for $\text{Row}(A)$.

$\{a_1, a_2, a_4\}$ forms a basis.

2. Find all values of c for which $\mathbf{v} = (2, 1, 4, -5)$ lies in $\text{Row}(A)$.

Question 3 — solution: part 1

1. $\operatorname{rref}(A^T) = \begin{pmatrix} 1 & 0 & 2 & 0 & -1 \\ 0 & 1 & c & 0 & 0 \\ 0 & 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$ (C_1 C_2 C_3 C_4 C_5)

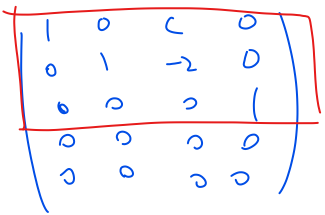
Since rref preserves linear combination of columns, so $\{a_3, a_4, a_5\}$ forms a basis of $\operatorname{Row}(A)$ $\iff$ $\{C_3, C_4, C_5\}$ forms a basis of $\operatorname{Col}(\operatorname{rref}(A^T))$ $\iff$ $\{C_3, C_4, C_5\}$ forms a basis of $\operatorname{Col}(A^T)$

So we look at $\begin{pmatrix} 2 & 0 & -1 \\ c & 0 & 0 \\ 0 & 1 & 3 \\ 0 & 0 & 0 \end{pmatrix}$

$\{C_2, C_4, C_5\}$ is l.i.

$\iff \det\begin{pmatrix} 2 & 0 & -1 \\ c & 0 & 0 \\ 0 & 1 & 3 \end{pmatrix} = -c \neq 0 \iff c \neq 0$ ※

Question 3 — solution: part 2

$\mathrm{rref}(A) =$ 

→ This gives basis of $\mathrm{Row}(A)$.

Question: For which c, $(2 \;\; 1 \;\; 4 \;\; -5) \in \mathrm{Span}\left\{ 2(1\ 0\ c\ 0),\ 1(0\ 1\ -2\ 0),\ -5(0\ 0\ 0\ 1) \right\}$?

$v \in \mathrm{Row}(A) \iff \exists\, x, y, z$ s.t.

$$v^t = \begin{pmatrix} 2 \\ 1 \\ 4 \\ -5 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ c & -2 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} x \\ y \\ cx - 2y \\ z \end{pmatrix} \quad (\Rightarrow \; x = 2,\ y = 1,\ z = -5.$$

$$\iff \begin{pmatrix} 2 \\ 1 \\ 4 \\ -5 \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \\ 2c - 2 \\ -5 \end{pmatrix} \iff 4 = 2c - 2 \iff c = 3$$

Next lesson

Next lesson (**Fri 13 Nov**), we finish the revision with

1. Chapter 5: Orthogonality, Projection, and Least Squares Solution, and

2. Chapter 6: Eigenanalysis.

Please bring the questions you would like to go through.